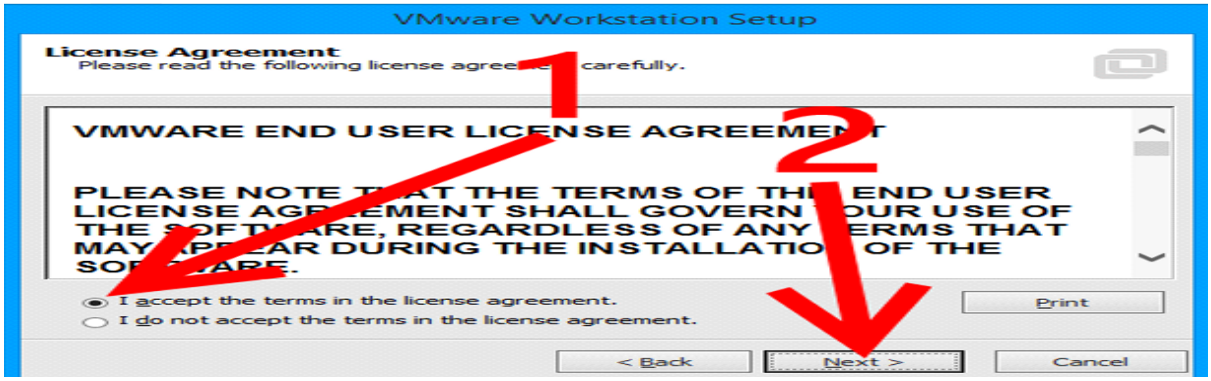
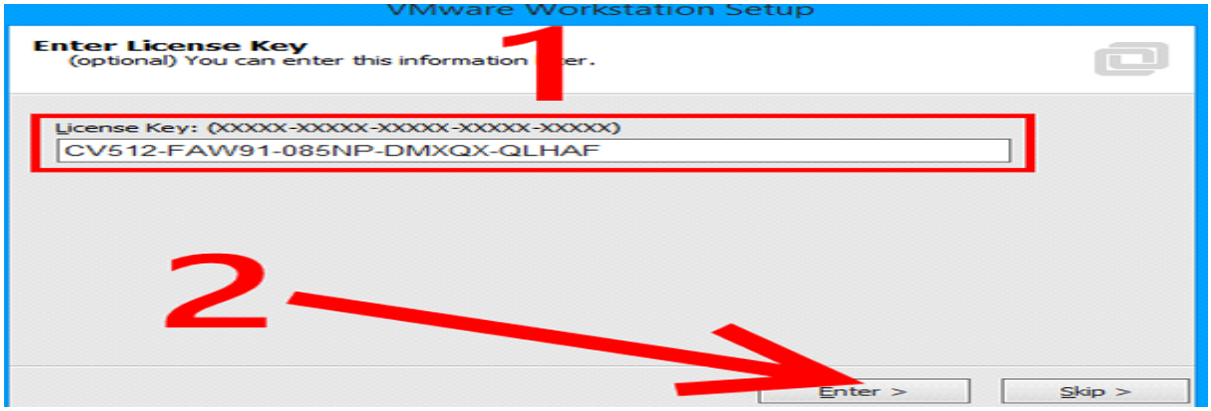
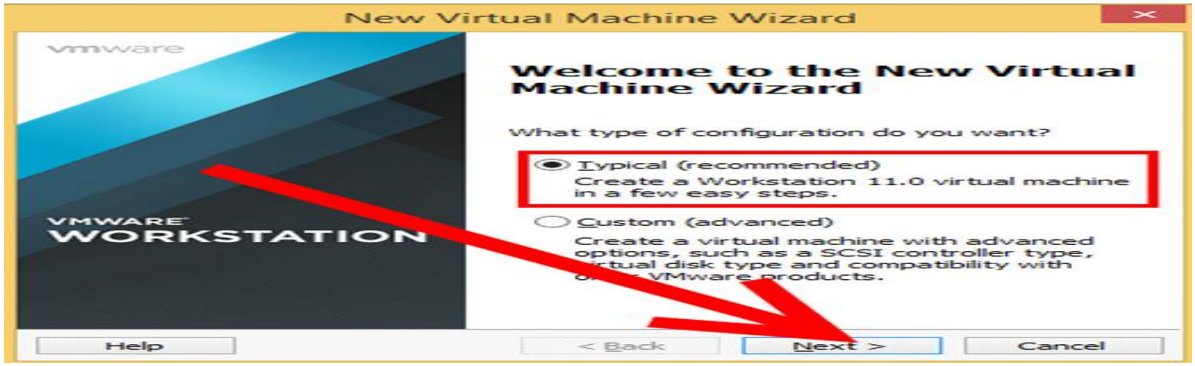
BIG DATA TECHNOLOGIES LAB

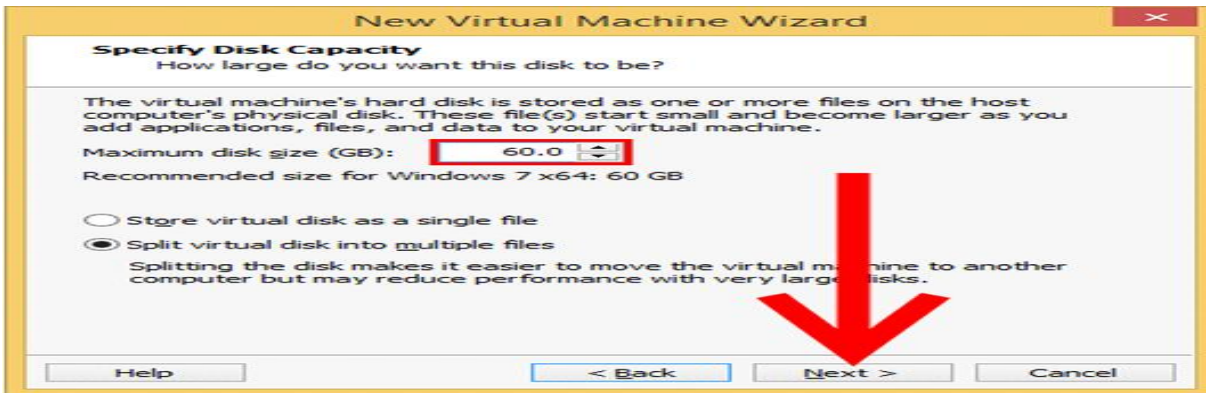
Week-1

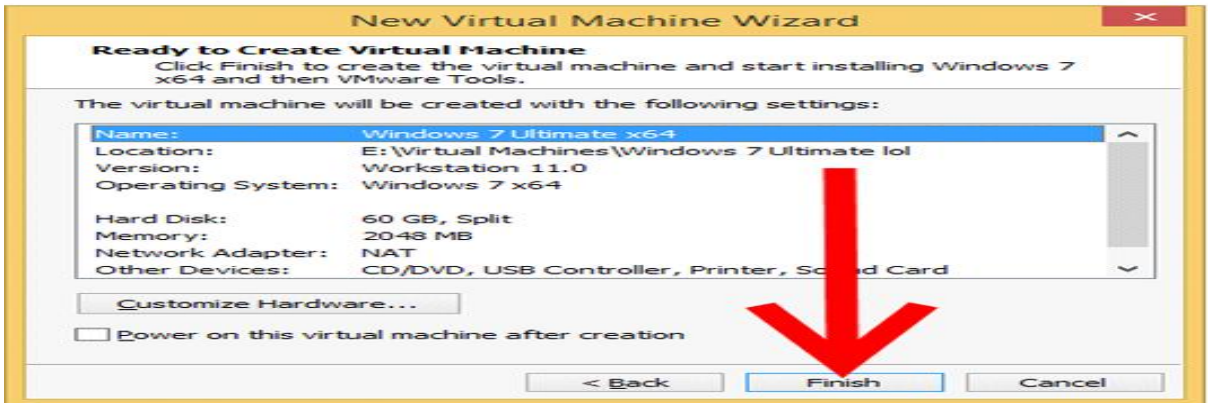
i)Perform setting and installing Vmware for Hadoop & Linux






New Virtual Machine Wizard
vmware
VMWARE WORKSTATION
Welcome to the New Virtual Machine Wizard
What type of configuration do you want?
Typical (recommended)
Create a Workstation 11.0 virtual machine in a few easy steps.
Custom (advanced)
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with other VMware products.
Help
< Back
Next >
Cancel


New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB): 60.0
Recommended size for Windows 7 x64: 60 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel


New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Windows 7 x64 and then VMware Tools.
The virtual machine will be created with the following settings:
Name: Windows 7 Ultimate x64
Location: E:\Virtual Machines\Windows 7 Ultimate lol
Version: Workstation 11.0
Operating System: Windows 7 x64
Hard Disk: 60 GB, Split
Memory: 2048 MB
Network Adapter: NAT
Other Devices: CD/DVD, USB Controller, Printer, Sound Card
Customize Hardware...
Power on this virtual machine after creation
< Back
Finish
Cancel

Windows XP Professional
File
Home
Share
View
Virtual Machines
Windows XP Professional
Search Windows XP...
Favorites
Homegroup
This PC
Network
Windows XP Professional.nvram
Windows XP Professional.vmdk
Windows XP Professional.vmsd
Windows XP Professional.vmx
Windows XP Professional.vmxf
Windows XP Professional-s001.vmdk
Windows XP Professional-s002.vmdk
Windows XP Professional-s003.vmdk

ii) Basic Linux Commands

i)mkdir

```
sr204@sr204-Veriton-M200-H510: ~/22R21A12J9
sr204@sr204-Veriton-M200-H510:~$ mkdir 22R21A12J9
sr204@sr204-Veriton-M200-H510:~$ ls
22R21A12J9  Desktop    Downloads  Pictures  snap       Videos
akhila      Documents  Music      Public    Templates
```

ii)man

```
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$ man man
man: can't resolve man7/groff_man.7
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$ S
```

iii) touch

```
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$ touch abcd
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$ cat>abcd
Iam from Mlrit
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$ cat abcd
Iam from Mlrit
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$ touch xyz
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$ cp abcd xyz
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$
```

iv)mv

```
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$ mv abcd xyz
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$ cat abcd
cat: abcd: No such file or directory
sr204@sr204-Veriton-M200-H510:~/22R21A12J9$ cat xyz
My name is akhila.
Iam studying 3rd year.
```

v) ls

```
prateekjangid@prateekjangid:~$ ls
Desktop  Documents  INFO    Music     Public  Templates
DETAIL   Downloads  LINUX   Pictures  snap    Videos
prateekjangid@prateekjangid:~$ rmdir LINUX INFO DETAIL
prateekjangid@prateekjangid:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
prateekjangid@prateekjangid:~$
```

vi) echo

```
raghvendra@raghvendra-Inspiron-15-3567:~$ echo "Geeks for Geeks"
Geeks for Geeks
raghvendra@raghvendra-Inspiron-15-3567:~$
```

vii) rm

```
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ rm a.txt
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ cat a.txt
cat: a.txt: No such file or directory
```

viii) mv

```
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ cat x.txt
Hi
Hello
I am from MLRIT

onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ mv x.txt a.txt
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ cat a.txt
Hi
Hello
I am from MLRIT
```

ix) cp

```
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ cat x.txt
Hi
Hello
I am from MLRIT

onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ cp x.txt z.txt
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ cat z.txt
Hi
Hello
I am from MLRIT
```

x) pwd

```
root@DESKTOP-[redacted]:/home/hp# passwd user1
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@DESKTOP-[redacted]:/home/hp#
```

iii) Run basic HDFS shell commands

mkdir

```
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -mkdir /geeks
19/01/31 10:53:43 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -ls /
19/01/31 10:53:56 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
Found 5 items
drwxr-xr-x   - suraj supergroup          0 2019-01-31 10:53 /geeks
-rw-r--r--   1 suraj supergroup   13965969 2019-01-31 00:13 /input
drwxr-xr-x   - suraj supergroup          0 2019-01-31 01:30 /output
drwx------   - suraj supergroup          0 2019-01-31 00:15 /tmp
drwxr-xr-x   - suraj supergroup          0 2019-01-30 23:44 /user
```

ls

```
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -ls /
19/01/31 10:35:11 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
Found 4 items
-rw-r--r--   1 suraj supergroup   13965969 2019-01-31 00:13 /input
drwxr-xr-x   - suraj supergroup          0 2019-01-31 01:30 /output
drwx------   - suraj supergroup          0 2019-01-31 00:15 /tmp
drwxr-xr-x   - suraj supergroup          0 2019-01-30 23:44 /user
```

touchz

```
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -touchz /geeks/myfile.txt
19/01/31 11:10:31 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -lsr /geeks
lsr: DEPRECATED: Please use 'ls -R' instead.
19/01/31 11:10:48 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
-rw-r--r--   1 suraj supergroup          0 2019-01-31 11:10 /geeks/myfile.txt
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$
```

BIG DATA TECHNOLOGIES LAB

Week-2

Implement the following file management tasks in Hadoop:

i)Adding files and directories ii) Retrieving files

iii) deleting files and directories

1)Adding files and directories

Mkdir

```
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -mkdir /geeks
19/01/31 10:53:43 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -ls /
19/01/31 10:53:56 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
Found 5 items
drwxr-xr-x   - suraj supergroup          0 2019-01-31 10:53 /geeks
-rw-r--r--   1 suraj supergroup   13965969 2019-01-31 00:13 /input
drwxr-xr-x   - suraj supergroup          0 2019-01-31 01:30 /output
drwx------   - suraj supergroup          0 2019-01-31 00:15 /tmp
drwxr-xr-x   - suraj supergroup          0 2019-01-30 23:44 /user
```

put

```
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -put ../Desktop/AI.txt /geeks
19/01/31 11:31:02 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -lsr /geeks
lsr: DEPRECATED: Please use 'ls -R' instead.
19/01/31 11:31:21 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
-rw-r--r--   1 suraj supergroup        205 2019-01-31 11:31 /geeks/AI.txt
-rw-r--r--   1 suraj supergroup          0 2019-01-31 11:10 /geeks/myfile.txt
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$
```

2)Retrieving files

Ls

```
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -ls /
19/01/31 10:35:11 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
Found 4 items
-rw-r--r--   1 suraj supergroup   13965969 2019-01-31 00:13 /input
drwxr-xr-x   - suraj supergroup          0 2019-01-31 01:30 /output
drwx------   - suraj supergroup          0 2019-01-31 00:15 /tmp
drwxr-xr-x   - suraj supergroup          0 2019-01-30 23:44 /user
```

get

```
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -get /geeks/myfile.txt  ../Desktop/hero
19/01/31 11:43:34 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ ls ../Desktop/hero
myfile.txt
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$
```

3) Deleting files and directories

rm

```
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -rmr /geeks_copied
rmr: DEPRECATED: Please use 'rm -r' instead.
19/01/31 12:58:40 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
19/01/31 12:58:42 INFO fs.TrashPolicyDefault: Namenode trash configuration: Deletion interval = 0 minutes, Emptier interval = 0 minutes.
Deleted /geeks_copied
suraj@suraj:~/hadoop-2.5.0-cdh5.3.2$ bin/hdfs dfs -ls /geeks_copied
19/01/31 12:59:35 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
```

BIG DATA TECHNOLOGIES LAB

Week-3

1) Develop a MapReduce program to calculate the frequency of a given word in a given file.

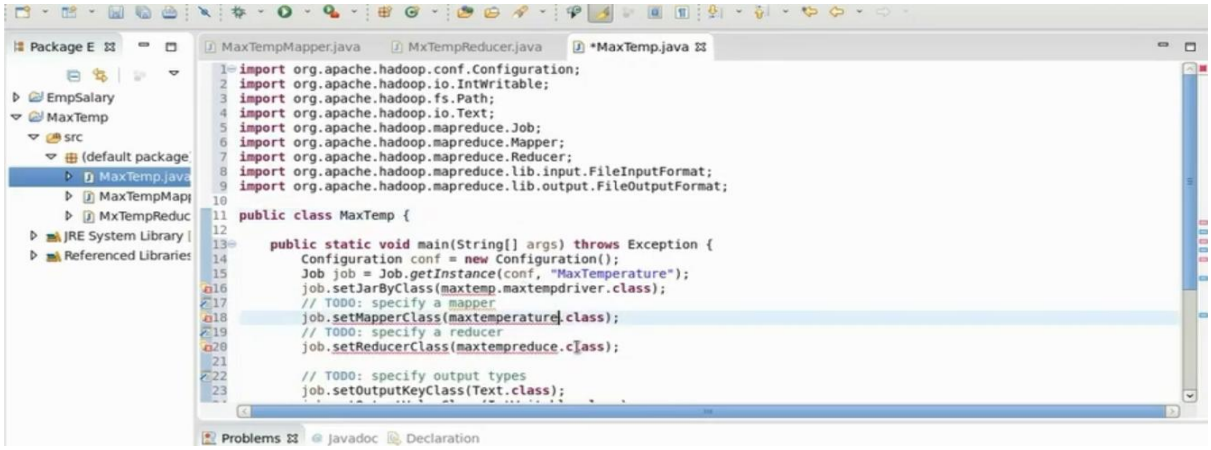
```
Last login: Mon Oct 10 00:37:23 on ttys000
→  ~ hadoop version
Hadoop 3.2.3
Source code repository https://github.com/apache/hadoop -r abe5358143720085498613d399be3bbf01e0f131
Compiled by ubuntu on 2022-03-20T01:18Z
Compiled with protoc 2.5.0
From source with checksum 39bb14faec14b3aa25388a6d7c345fe8
This command was run using /Users/arjungautam/hadoop-3.2.3/share/hadoop/common/hadoop-common-3.2.3.jar
→  ~
```

```
C:\hadoop\sbin>hadoop fs -mkdir /input_dir

C:\hadoop\sbin>hadoop fs -put C:/Users/JAWAD/Desktop/Hammad/HADOOPINSTALLATION/data.txt /input_dir

C:\hadoop\sbin>hadoop fs -ls /input_dir/
Found 1 items
-rw-r--r--   1 JAWAD supergroup        111 2022-06-07 00:27 /input_dir/data.txt
```

```
C:\hadoop\sbin>hadoop fs -cat /input_dir/data.txt
Pakistan
India
China
Bangladesh
Pakistan
Iran
America
India
Iran
America
China
Pakistan
Iraq
China
```

```
C:\hadoop\sbin>hadoop fs -cat /output_dir/*
America 2
Bangladesh      1
China   3
India   2
Iran    2
Iraq    1
Pakistan        3
```

2) Develop a MapReduce program to find the maximum temperature in each year.




```
        Total megabyte-milliseconds taken by all reduce tasks=3511296
    Map-Reduce Framework
        Map input records=5
        Map output records=5
        Map output bytes=45
        Map output materialized bytes=61
        Input split bytes=119
        Combine input records=0
        Combine output records=0
        Reduce input groups=2
        Reduce shuffle bytes=61
        Reduce input records=5
        Reduce output records=2
        Spilled Records=10
        Shuffled Maps =1
        Failed Shuffles=0
        Merged Map outputs=1
        GC time elapsed (ms)=96
        CPU time spent (ms)=840
        Physical memory (bytes) snapshot=389406720
        Virtual memory (bytes) snapshot=3015147520
        Total committed heap usage (bytes)=292163584
    Shuffle Errors
        BAD_ID=0
        CONNECTION=0
        IO_ERROR=0
        WRONG_LENGTH=0
        WRONG_MAP=0
        WRONG_REDUCE=0
    File Input Format Counters
        Bytes Read=530
    File Output Format Counters
```

Output:

```
            Bytes Written=17
[cloudera@quickstart Desktop]$ hadoop dfs -cat /maxoutput/part-r-00000
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.

1949    111
1950    22
```

BIG DATA TECHNOLOGIES LAB

Week-4

1) Design MapReduce algorithms to take a very large file of integers and produce as output:

i) The largest integer

ii) The average of all the integers.

iii) The same set of integers, but with each integer appearing only once.

iv) The count of the number of distinct integers in the input

```
// File: IntegersAllInOne.java
// Modes: max | avg | distinct | countdistinct
// Usage:
//   hadoop jar integers.jar IntegersAllInOne <mode> <input_path> <output_path>
//
// Example:
//   hadoop jar integers.jar IntegersAllInOne max /input/numbers.txt /out_max

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class IntegersAllInOne {
```

```
    // ---------- (i) Largest integer ----------
    public static class MaxMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
        private static final Text K = new Text("k");
        @Override
        protected void map(LongWritable key, Text value, Context ctx) throws IOException, Inte
            String s = value.toString().trim();
            if (s.isEmpty()) return;
            ctx.write(K, new IntWritable(Integer.parseInt(s)));
        }
    }
    public static class MaxReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
        @Override
        protected void reduce(Text key, Iterable<IntWritable> vals, Context ctx) throws IOExce
            int max = Integer.MIN_VALUE;
            for (IntWritable v : vals) max = Math.max(max, v.get());
            ctx.write(new Text("Largest"), new IntWritable(max));
        }
    }

    // ---------- (ii) Average of integers ----------
    public static class AvgMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
        private static final Text K = new Text("k");
        @Override
        protected void map(LongWritable key, Text value, Context ctx) throws IOException, Inte
            String s = value.toString().trim();
            if (s.isEmpty()) return;
            ctx.write(K, new IntWritable(Integer.parseInt(s)));
        }
    }
    public static class AvgReducer extends Reducer<Text, IntWritable, Text, Text> {
        @Override
        protected void reduce(Text key, Iterable<IntWritable> vals, Context ctx) throws IOExce
            long sum = 0; long count = 0;
            for (IntWritable v : vals) { sum += v.get(); count++; }
            double avg = (count == 0) ? 0.0 : (sum * 1.0) / count;
            ctx.write(new Text("Average"), new Text(String.valueOf(avg)));
```

```
    // ---------- Driver ----------
    public static void main(String[] args) throws Exception {
        if (args.length != 3) {
            System.err.println("Usage: IntegersAllInOne <mode:max|avg|distinct|countdistinct
            System.exit(2);
        }
        String mode = args[0];
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Integers-" + mode);
        job.setJarByClass(IntegersAllInOne.class);

        FileInputFormat.addInputPath(job, new Path(args[1]));
        FileOutputFormat.setOutputPath(job, new Path(args[2]));

        switch (mode.toLowerCase()) {
            case "max":
                job.setMapperClass(MaxMapper.class);
                job.setReducerClass(MaxReducer.class);
                job.setMapOutputKeyClass(Text.class);
                job.setMapOutputValueClass(IntWritable.class);
```

Numbers.txt

```
5
7
5
2
9
2
```

Complile & Run :

```
# Put input in HDFS
hdfs dfs -mkdir -p /input
hdfs dfs -put -f numbers.txt /input/

# (i) Largest
hdfs dfs -rm -r -f /out_max
hadoop jar integers.jar IntegersAllInOne max /input/numbers.txt /out_max

# (ii) Average
hdfs dfs -rm -r -f /out_avg
hadoop jar integers.jar IntegersAllInOne avg /input/numbers.txt /out_avg

# (iii) Distinct set
hdfs dfs -rm -r -f /out_distinct
hadoop jar integers.jar IntegersAllInOne distinct /input/numbers.txt /out_distinct

# (iv) Count of distinct
hdfs dfs -rm -r -f /out_countdistinct
hadoop jar integers.jar IntegersAllInOne countdistinct /input/numbers.txt /out_countdistinct
```

```
$ hdfs dfs -cat /out_max/part-r-00000
Largest■9
```

```
$ hdfs dfs -cat /out_avg/part-r-00000
Average■5.0
```

```
$ hdfs dfs -cat /out_distinct/part-r-00000
2
5
7
9
```

```
$ hdfs dfs -cat /out_countdistinct/part-r-00000
DistinctCount■4
```

BIG DATA TECHNOLOGIES LAB

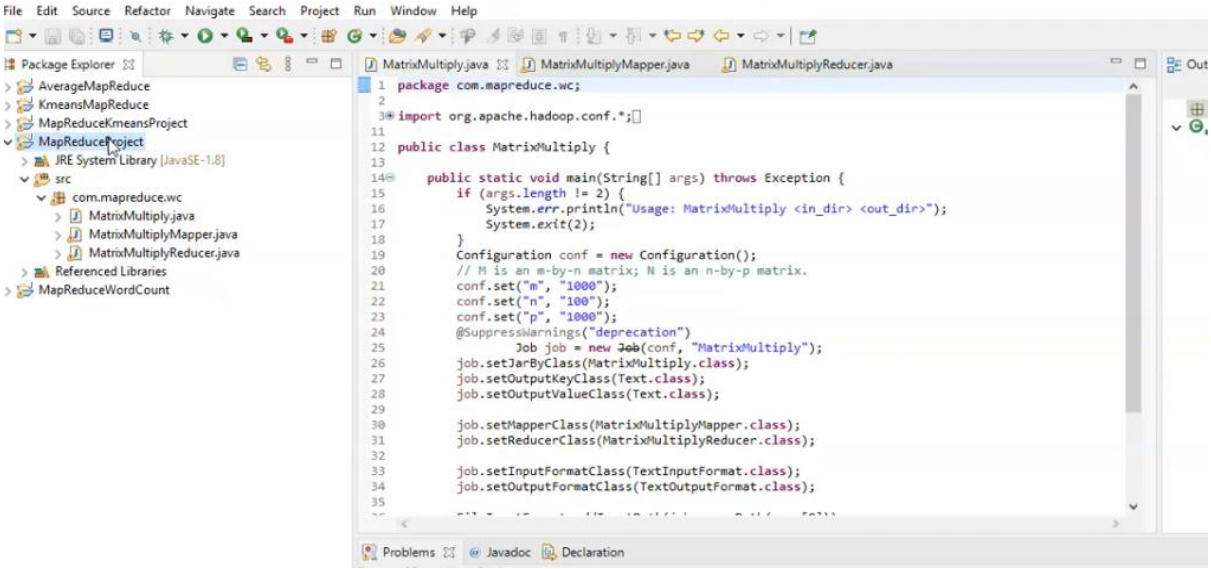
Week-5

1) Implement Matrix Multiplication on Hadoop using map reduce

```
G:\hadoop\sbin>hadoop fs -mkdir /input_matrix

G:\hadoop\sbin>hadoop fs -put G:\Hadoop_Experiments\MatrixMultiply_M.txt /input_matrix

G:\hadoop\sbin>hadoop fs -ls /input_matrix
Found 1 items
-rw-r--r--   1 hp supergroup         34 2021-04-06 00:07 /input_matrix/MatrixMultiply_M.txt
```

```
G:\hadoop\sbin>hadoop dfs -cat /input_matrix/MatrixMultiply_M.txt
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
M,0,0,1
M,0,1,2
M,1,0,3
M,1,1,4
G:\hadoop\sbin>
```

```
G:\hadoop\sbin>hadoop fs -put G:\Hadoop_Experiments\MatrixMultiply_N.txt /input_matrix

G:\hadoop\sbin>hadoop dfs -cat /input_matrix/MatrixMultiply_N.txt
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
N,0,0,
N,0,1,6
N,1,0,7
N,1,1,8
G:\hadoop\sbin>
```



```
N,1,1,8
G:\hadoop\sbin>hadoop jar G:\Hadoop_Experiments\MatrixMultiply.jar com.mapreduce.wc/MatrixMultiply /input
ut_matrix
2021-04-06 00:12:38,265 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2021-04-06 00:12:39,503 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not perfor
the Tool interface and execute your application with ToolRunner to remedy this.
2021-04-06 00:12:39,583 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoo
hp/.staging/job_1617647692087_0001
2021-04-06 00:12:40,074 INFO input.FileInputFormat: Total input files to process : 2
2021-04-06 00:12:40,518 INFO mapreduce.JobSubmitter: number of splits:2
2021-04-06 00:12:40,753 INFO Configuration.deprecation: yarn.resourcemanager.system-metrics-publisher.ena
ted. Instead, use yarn.system-metrics-publisher.enabled
2021-04-06 00:12:41,587 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1617647692087_0001
2021-04-06 00:12:41,589 INFO mapreduce.JobSubmitter: Executing with tokens: []
```

```
                Peak Reduce Virtual memory (bytes)=258981888
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=68
        File Output Format Counters
                Bytes Written=36
```

Output:

```
G:\hadoop\sbin>hadoop dfs -cat /output_matrix/*
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
0,0,19.0
0,1,22.0
1,0,43.0
1,1,50.0
```

BIG DATA TECHNOLOGIES LAB

Week-6

Sh

```
grunt> sh shell command parameters
```

```
grunt> sh ls

pig
pig_1444799121955.log
pig.cmd
pig.py
```

fs

```
grunt> fs ls

Found 3 items
drwxrwxrwx   - Hadoop supergroup          0 2015-09-08 14:13 Hbase
drwxr-xr-x   - Hadoop supergroup          0 2015-09-09 14:52 seqgen_data
drwxr-xr-x   - Hadoop supergroup          0 2015-09-08 11:30 twitter_data
```

```
grunt> clear
```

clear

```
grunt> clear
```

help

```
grunt> help
```

```
Commands: <pig latin statement>; - See the PigLatin manual for details:
http://hadoop.apache.org/pig

File system commands:fs <fs arguments> - Equivalent to Hadoop dfs  command:
http://hadoop.apache.org/common/docs/current/hdfs_shell.html

Diagnostic Commands:describe <alias>[::<alias] - Show the schema for the alias.
Inner aliases can be described as A::B.
    explain [-script <pigscript>] [-out <path>] [-brief] [-dot|-xml]
       [-param <param_name>=<pCram_value>]
       [-param_file <file_name>] [<alias>] -
       Show the execution plan to compute the alias or for entire script.
       -script - Explain the entire script.
```

quit

```
grunt> quit
```

```
grunt> exec [param param_name = param_value] [param_file file_name] [script]
```

exec

```
grunt> exec [param param_name = param_value] [param_file file_name] [script]
```

kill

```
grunt> kill JobId
```

```
grunt> kill Id_0055
```

dump

```
grunt> Dump;

(1,Rajiv,Hyderabad)
(2,siddarth,Kolkata)
(3,Rajesh,Delhi)
```

2) Write Pig Latin scripts to sort, group, join, project, and filter your data.

students.txt

```
001, Rajiv, Reddy, 21, 9848022337, Hyderabad
002, Siddarth, Battacharya, 22, 9848022338, Kolkata
003, Rajesh, Khanna, 22, 9848022339, Delhi
004, Preethi, Agarwal, 21,9848022330, Pune
005, Trupthi, Mohanthy, 23,9848022336, Bhubaneshwar
006, Archana, Mishra, 23,9848022335, Chennai
007, Komal, Nayak, 24, 9848022334, Trivendram
008, Bharathi, Nambiayar, 24,9848022333, Chennai
--Press ctrl+W to save the file
```

Sort :

```
sorted_students = ORDER student_data BY age DESC;
DUMP sorted_students;
grunt> sorted_students = ORDER student_data BY age DESC;
grunt> DUMP sorted_students;
```

✅ **Orders students from oldest to youngest.**

```
(8,Bharathi,Nambiayar,24,9450987125,Chennai)
(7,Komal,Nayak,24,9856780237,Trivendram)
(6,Archana,Mishra,23,9856783451,Chennai)
(5,Trupthi,Mohanthy,23,9812376590,Bhubaneshwar)
(3,Rajesh,Khanna,22,9894567834,Delhi)
(2,Siddharth,Battacharya,22,9845673290,Kolkata)
(4,Preethi,Agarwal,21,9847681234,Pune)
(1,Rajiv,Reddy,21,9876543210,Hyderabad)
```

Group :

```
grouped_students = GROUP student_data BY city;
DUMP grouped_students;
grunt> grouped_students = GROUP student_data BY city;
grunt> DUMP grouped_students;
```

✅ **Groups students based on their city.**

```
(Pune,{(4,Preethi,Agarwal,21,9847681234,Pune)})
(Delhi,{(3,Rajesh,Khanna,22,9894567834,Delhi)})
(Chennai,{(8,Bharathi,Nambiayar,24,9450987125,Chennai),(6,Archana,Mishra,23,9856
783451,Chennai)})
(Kolkata,{(2,Siddharth,Battacharya,22,9845673290,Kolkata)})
(Hyderabad,{(1,Rajiv,Reddy,21,9876543210,Hyderabad)})
(Trivendram,{(7,Komal,Nayak,24,9856780237,Trivendram)})
(Bhubaneshwar,{(5,Trupthi,Mohanthy,23,9812376590,Bhubaneshwar)})
```

Project :

```
projected_students = FOREACH student_data GENERATE firstname, lastname, city;
DUMP projected_students;
grunt> projected_students = FOREACH student_data GENERATE firstname, lastname, c
ity;
grunt> DUMP projected_students;
```

✅ **Displays only selected fields.**

```
(Rajiv,Reddy,Hyderabad)
(Siddharth,Battacharya,Kolkata)
(Rajesh,Khanna,Delhi)
(Preethi,Agarwal,Pune)
(Trupthi,Mohanthy,Bhubaneshwar)
(Archana,Mishra,Chennai)
(Komal,Nayak,Trivendram)
(Bharathi,Nambiayar,Chennai)
```

Filter:

```
filtered_students = FILTER student_data BY age > 22;
DUMP filtered_students;
grunt> filtered_students = FILTER student_data BY age > 22;
grunt> DUMP filtered_students;
```

✅ **Filters students who are older than 22 years.**

```
(5,Trupthi,Mohanthy,23,9812376590,Bhubaneshwar)
(6,Archana,Mishra,23,9856783451,Chennai)
(7,Komal,Nayak,24,9856780237,Trivendram)
(8,Bharathi,Nambiayar,24,9450987125,Chennai)
```